

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_b43bcbe7-b94\agent-ae0b1a0e4d16c3ac4.jsonl`  
- SHA-256: `887d1c2f76f4a6f5867173faa7c4126fdf749033cad894bbebae0ce7f9cf84a7`  
- Source modified: `2026-07-02T03:20:00+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_b43bcbe7-b94`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T03:16:59.072Z)

You are an adversarial verifier in a tech-debt audit of the multi-repo workspace at C:/Users/avidu/Projects/khelsutra-guru (repos: badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport; non-git dirs 'Annotation Setup', khelsutra-screencast). Today is 2026-07-02. All repos are at origin head.
Default stance: REFUTE. A round-1 finder claimed (UNVERIFIED):
REPO: badminton-highlight-indexer
CLAIM: Upload retention sweep tool exists but nothing schedules it ? unbounded upload growth + unenforced 7-day privacy retention on any hosted deployment (#301)
INVESTIGATION HINTS: Find the sweep tool (tools/ or backend/), check for any scheduler wiring (nightly.yml, deploy scripts, startup hooks); gh issue view 301.
VERIFY against the CURRENT working tree: (1) reproduce the defect/debt from actual code at file:line; (2) check git log for a fix that already landed; (3) construct the concrete failure scenario or downgrade severity; (4) propose the correctly-scoped fix with effort S/M/L; (5) flag OWNER_GATED if it needs owner decisions (real cloud spend, product/strategy, counsel, Cloudflare dashboard).
HARD RULES: READ-ONLY ? no file modifications, no installs (pip download/index checks and WebFetch to pypi.org are OK), no test-suite runs. gh read-only OK. sqlite reads OK. CONFIRMED requires you personally observed the defect in current code/data; cite file:line for every claim.

2. user (2026-07-02T03:17:07.025Z)

output

3. user (2026-07-02T03:17:08.445Z)

No files found

4. user (2026-07-02T03:17:14.069Z)

output

```
---  
Annotation Setup  
badminton-highlight-indexer  
khelsutra  
khelsutra-screencast  
rally-annotator  
sports-data-collector  
sports-obsreport
```

5. user (2026-07-02T03:17:21.320Z)

.

```
..
.claude
output
---
CLAUDE.md
README.md
__pycache__
artifacts
backend
backend.db
config.json
deploy
docs
eval_baselines
indexer.db
mypy.ini
output
pyproject.toml
pytest.ini
rally_index.db
requirements.txt
ruff.toml
run_all_tests.py
run_eval.py
run_regression.py
scratch
scripts
tests
tools
training
```

6. user (2026-07-02T03:17:32.948Z)

Found 80 files

```
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_wasb_runner.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\CODE_MAP.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\NEXT_STEPS.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\deploy\gcp\nightly_regression\launch.sh
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\deploy\gcp\create_gpu_vm.sh
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\deploy\gcp\nightly_regression\create_nightly_bucket.sh
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\deploy\gcp\nightly_regression\README.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\main.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\CLAUDE.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\README.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\.github\workflows\ci.yml
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_rally_seg_eval.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\NIGHTLY_ENGINE_REGRESSION.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tools\cleanup_caches.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_tolerance_metrics.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_segmentation_metrics.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_server_entrypoint.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_provider_gemini.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_reingest_safety.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_p2b_install.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_per_user_eval.py
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\tests\test_flywheel.py
```

C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\tests\test_eval_partial_labels.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\tests\test_async_jobs.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\tests\test_base.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\tests\test_cleanup_caches.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\scripts\uninstall.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\lovo_resweep.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\CODE_CLEANUP_AUDIT_2026-06-24.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\pyproject.toml
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\run_eval.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\scripts\install.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\utils\single_instance.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\providers\gemini.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\pipeline\detectors\person_detector.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\segmentation_metrics.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\tolerance_metrics.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\flywheel.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\infrastructure\database.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\per_user_eval.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\metrics.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\eval_partial_labels.py
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\.github\workflows\nightly.yml
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\README.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\mypy.ini
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\archives\quality-iterations\README.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\archives\past_projects\DECOUPLED_COMPUTE\06-RISKS-AND-OPEN-QUESTIONS.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\archives\PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\REAL_FOOTAGE_VALIDATION_REPORT.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\R1_MILESTONE_LAUNCH_REPORT.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\R2_EVAL_METRIC_DESIGN.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\RALLY_DETECTION_QUALITY_REPORT.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\PACKAGING_PLAN.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\COMPUTE_DECOUPLED_SERVING\README.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\BACKLOG.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\CLOUD_GPU_DRYRUN_GUIDE.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\ui\assets\index-ukuS9wxa.js
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\VIDEO_LOCALITY_MODEL.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\docs\STORAGE_SHARING_MODEL.md
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\

indexer\docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\EVALUATION.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\EXECUTION_POLICY.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\ANALYZERS_AND_RECONCILERS.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\tests\test_import_smoke.py
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\research\AUDIO_E1_FINDINGS.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\MONETIZATION_AUDIT.md

7. user (2026-07-02T03:17:33.861Z)

e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
aa915a7 deploy/gcp: centralize project + bucket names in one sourced buckets.env (#410)
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)

master...origin/master

8. user (2026-07-02T03:17:47.019Z)

/c/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/deploy:

cloudrun
digitalocean
gcp

/c/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/scripts:

README.md
__pycache__
bundle_ui.sh
doctor.py
install.py
nightly_email.py
nightly_reelcount.py
nightly_regression.ps1
nightly_regression.py
register_nightly_task.ps1
run_phase3_eval.py
run_process_and_stitch.py
uninstall.py
verify_compute_path.py

/c/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/tools:

__init__.py
__pycache__
check_md_links.py
cleanup_caches.py
generate_highlights.py
leak_scan.py

9. user (2026-07-02T03:17:49.231Z)

```
title: T11 follow-up: run the upload retention sweep as an automated scheduled job (hosted
alpha)
state: OPEN
author: avidullu (Avi)
labels: enhancement
comments: 0
assignees:
projects:
milestone:
number: 301
--
```

Context

The T11 **upload retention sweep** landed in #299 (merged, `cb7394b``): `tools/cleanup_caches.py`` was extended to purge ? past a retention window ? uploaded source videos and orphaned `.partial-<id>`` files in `security.upload_dir`` (default `output/uploads``), alongside the existing regenerable-cache cleanup. It is **dry-run by default**, keeps reels/DB/CSVs durable, and emits a machine-readable plan via `--json``.

That PR delivered the **tool**. What it did **not** do is run it **unattended**: today an operator must manually run `python -m tools.cleanup_caches --apply`` on the box. Before/while non-owner (alpha) users upload footage, the retention window should be enforced automatically without a human in the loop.

Goal

Wire the existing sweep into an **automated scheduled job** so stale uploads are purged on the owner-set retention window (uploads **7d**; reels kept durable) without manual intervention, with loud structured logging captured to observability.

Do not re-implement the sweep ? schedule the existing tool. The `--json`` output (`{windows, uploads_dir, reclaimable_bytes, purge[], by_category[]}``) is already shaped for a wrapper/log sink.

What's needed

- A scheduler appropriate to the deployment:
 - **Single-box alpha** (engine on one host): a host **cron** or **systemd-timer** entry that runs the sweep daily.
 - **Cloud Run serving path**: a **Cloud Scheduler** ? **job** invocation (or a sidecar), consistent with `COMPUTE_DECOUPLED_SERVING/``.
- The scheduled invocation runs `--apply`` with `--uploads-retention-days 7`` and `--uploads-dir`` matching `security.upload_dir`` (consider resolving from config rather than hardcoding the path).
- **Default-OFF / opt-in** so the single-user local install is never surprised by an unattended deletion job.
- **Observability**: capture the `--json`` plan + the per-item `PURGED/FAILED`` lines to logs each run (counts + bytes freed + the active windows), per the launch quality bar (loud, structured, correlation id).
- **Document** it in the deploy runbook (engine `deploy/`` and/or `khelsutra/deploy/DEPLOY_ALPHA.md``).

Design considerations / open questions

- **In-flight safety**: the retention window already protects recent `.partial-<id>`` files (an in-flight chunked upload < window is never swept). Confirm the cadence + window leave a comfortable margin over the max upload duration.
- **Active-job safety**: a **finished** upload may still be referenced by an in-progress `/api/process`` job. Decide whether the scheduled apply should additionally skip uploads referenced by a non-terminal row in the `jobs`` table (defense-in-depth beyond the time window). The window alone is probably sufficient for 7d, but worth a deliberate call.
- **Per-user subdirs (PR-3)**: the sweep currently leaves an unknown upload **subdir** untouched

(`classify` returns `upload subdir` = keep). When identity scoping lands, the job/tool will need to recurse per-user.

- **Apply guard:** since the scheduled job runs `--apply`, gate it behind an explicit config flag + keep the conservative window so a misconfiguration can't nuke fresh uploads.

Acceptance criteria

- [] A scheduled mechanism runs `tools/cleanup\_caches.py` on a cadence (e.g. daily) in the hosted/alpha deployment.
- [] It applies the owner-set window (uploads 7d, reels durable) and writes the structured `--json` plan + outcome to logs/observability.
- [] Default-OFF / opt-in; the local single-user path is unaffected.
- [] In-flight + active-job safety is reasoned about and documented (window margin, optional jobs-table check).
- [] The deploy runbook documents how to enable + verify it.
- [] A test/smoke confirms the wrapper invokes the tool with the right flags (the tool's own logic is already unit-tested).

References

- PR #299 (the tool) · `tools/cleanup\_caches.py` · `tests/test\_cleanup\_caches.py`
- `backend/api/uploads.py` (where uploads land) · `backend/config/models.py`
`SecurityConfig.upload\_dir`
- Cache-hygiene policy (retention table) · `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` §9 (T11, khelsutra repo) · the `ui-driven-cloud-serving-gap` thread

Filed as the scheduling follow-up to #299; the sweep tool itself is done and merged.

10. user (2026-07-02T03:17:59.163Z)

Found 7 files

C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\CODE_MAP.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\backend\pipeline\detectors\base.py
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\README.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\tools\cleanup_caches.py
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\tests\test_cleanup_caches.py
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\archives\PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md
C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\PER_VIDEO_WORKSPACE.md

11. user (2026-07-02T03:18:00.682Z)

aa915a7 deploy/gcp: centralize project + bucket names in one sourced buckets.env (#410)
b6214e5 chore: ruff format the whole repo (259 files) (#387)
9972057 ci: add Tier-1 nightly quality-regression workflow (#381)
cd5b3d2 feat(01/P11): Gemini remote-file GC ? startup orphan sweep (#380)
df3b225 docs(nightly): one-stop operations manual for the nightly engine-regression (#325)
48f9d20 feat(nightly): make the Cloud Run scheduler path work E2E + turnkey email/trigger (#320)
fa2342c feat(nightly): E2E engine reel-count regression on a GCP GPU VM (emailed) + cost (#315)
f8b2c3b feat(pkg): P2b-install ? batteries-included install + truly-local 'motion' first-run default (gate 08) (#310)
cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state (default-OFF) (#298)
7835f34 feat(gcp): always-install Ops Agent on GPU VMs + a cloud-GPU dry-run guide (#291)
9736a3b docs: re-ground stale doc claims to the current code (multi-agent audit) (#242)
5409fc4 refactor(api): extract job-worker drain/wake loop from main into backend.api.job_worker (#234)
67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13 (#204)
55e4c89 docs(archive): LOV0 threshold re-sweep at n=13 ? milestone thresholds hold (#214)
2eaccd5 feat(eval): nightly rally-quality regression tripwire (golden corpus, CPU) (#202)
7e8f88f feat(eval): R2 ? task-faithful tolerance-match rally metric (augments tIoU) (#199)

cf8973e docs(eval): R2 design ? task-faithful tolerance-match rally metric (FOR REVIEW) (#198)
033e26e Merge pull request #174 from avidullu/docs/consistency-sweep-173
e69e23d docs: fix doc drift across 16 docs (#173)

12. user (2026-07-02T03:18:11.448Z)

```
1483:# sequential-part logic, and abort cleanup ? is unchanged.
1808:def _maybe_sweep_gemini_orphans(config) -> None:
1809:     """O1/P11: best-effort sweep of Gemini File-API uploads orphaned by a previous crashed
run.
1811:     a failed sweep is logged and swallowed. The Gemini client is built with an HTTP timeout
so
1823:         deleted = provider.cleanup_orphaned_files()
1826:         "[STARTUP] Gemini orphan sweep reclaimed %d remote file(s).", deleted
1829:         logger.warning("[STARTUP] Gemini orphan sweep skipped (non-fatal): %s", e)
1999:     # P0e: refuse a SECOND instance against the same DB BEFORE the destructive stale-job
sweep below.
2000:     # The job worker is in-process, so a concurrent process opening this DB would sweep
THIS one's
2036:     # O1/P11: sweep Gemini File-API uploads orphaned by a hard process death (the per-call
2039:     _maybe_sweep_gemini_orphans(global_config)
2183:     # instead of the Optional module-level globals (CODE_CLEANUP_AUDIT P5 / D2).
```

13. user (2026-07-02T03:18:12.771Z)

```
deploy/gcp/nightly_regression/create_nightly_bucket.sh:13:# RALLY_NIGHTLY_TTL_DAYS=14 bash
.../create_nightly_bucket.sh # custom retention
deploy/gcp/nightly_regression/README.md:55:| **Report retention** | 30 days | the nightly
bucket's lifecycle TTL ? `RALLY_NIGHTLY_TTL_DAYS` then re-run `create_nightly_bucket.sh` |
```

14. user (2026-07-02T03:18:22.378Z)

""Cache-hygiene + retention sweep for the GPU highlight pipeline.

Scans the regenerable intermediates the pipeline leaves behind ? in the WSL stage
dir (`~/clips`), the Windows `output/` tree, and the upload landing zone
(`security.upload\_dir`, default `output/uploads`) ? classifies each entry as
purge (regenerable: frame caches, staged video copies, handoff clips, temp dirs;
and, past a retention window, uploaded source videos) or **keep** (durable: trajectory
CSVs, the SQLite DB, final reels) and reports the reclaimable space.

Retention sweep (T11 ? pre-alpha-exposure): uploaded source videos in the upload
dir are a *privacy*-driven purge, not a *disk*-driven one ? once a tester's footage is
older than `--uploads-retention-days` (default 7) it is swept, along with the orphaned
`.partial-<id>` temp files aborted uploads leave behind. Reels / DB / CSVs stay durable
per the cache-hygiene policy (the owner's call, 2026-06-21).

DRY-RUN BY DEFAULT ? it only *shows* what it would delete. Pass `--apply` to delete.

```
python -m tools.cleanup_caches # report only (safe)
python -m tools.cleanup_caches --apply # delete purge candidates
python -m tools.cleanup_caches --keep-video 89605e80ed01 # protect one video's cache
python -m tools.cleanup_caches --older-than 7 # only purge regenerable caches >7d old
python -m tools.cleanup_caches --uploads-retention-days 3 # sweep uploads >3d old
python -m tools.cleanup_caches --include-wasbcache # also drop detector resume caches
python -m tools.cleanup_caches --json # machine-readable plan (for a cron/job
wrapper)
```

Pairs with the pipeline's own self-cleaning (indexing.{wasb,tracknet}.keep_frames=false,
which deletes frames after a SUCCESSFUL run); this tool reclaims what failed/old runs
and pre-self-cleaning videos left behind, and enforces the upload retention window before
non-owner (alpha) exposure. Stdlib only; safe to run anywhere (the WSL scan is skipped
automatically when `wsl` is unavailable). `--uploads-dir` must match `security.upload\_dir`
(its default matches the config default, `output/uploads`).

```
"""
```

```
from __future__ import annotations
```

```
import argparse
```

```
import os
```

```
import shutil
```

```
import subprocess
```

```
import sys
```

```
from dataclasses import dataclass
```

```
from typing import Dict, List, Optional, Tuple
```

```
---- classification (pure; unit-tested) ----- #
```

(category, regenerable, default_purge). default_purge=True ? deleted unless a flag protects it. wasbcache is regenerable but default_purge=False (it enables resume and is tiny) ? opt in with --include-wasbcache.

```
_KEEP = ("keep", False, False)
```

```
def classify(name: str, is_dir: bool, location: str) -> Tuple[str, bool, bool]:
```

```
    """Classify a cache entry by name. Returns (category, regenerable, default_purge)."""
```

```
    low = name.lower()
```

```
    if location == "uploads":
```

```
        # The upload landing zone (security.upload_dir). Uploaded source videos and the
```

```
        # orphaned `.partial-<id>` temp files aborted uploads leave behind are both swept
```

```
        # past the retention window (default_purge=True; held by --uploads-retention-days
```

```
        # in plan_deletions). Anything else here ? e.g. a future per-user subdir (PR-3) ? is
```

```
        # left untouched, since we don't recurse into unknown upload trees.
```

```
    if is_dir:
```

```
        return ("upload subdir", False, False)
```

```
    if low.startswith(".partial-"):
```

```
        return ("partial upload", True, True) # aborted/in-flight chunk file
```

```
    if low.endswith((".mp4", ".avi", ".mov", ".mkv")):
```

```
        return ("uploaded source", True, True) # tester footage ? privacy retention
```

```
    return ("other", False, False)
```

```
    if is_dir:
```

```
        if low.endswith("_frames"):
```

```
            return ("frame cache", True, True)
```

```
        if low.endswith("_wasbcache"):
```

```
            return (
```

```
                "detector resume cache",
```

```
                True,
```

```
                False,
```

```
            ) # opt-in via --include-wasbcache
```

```
    if low.startswith("chunks_") and low.endswith("_handoff"):
```

```
        return ("ai handoff clips", True, True)
```

```
    if low.startswith("tmp"):
```

```
        return ("temp dir", True, True)
```

```
=====
```

```
12:older than ``--uploads-retention-days`` (default 7) it is swept, along with the orphaned
```

```
16:DRY-RUN BY DEFAULT ? it only *shows* what it would delete. Pass ``--apply`` to delete.
```

```
22: python -m tools.cleanup_caches --uploads-retention-days 3 # sweep uploads >3d old
```

```
58: # past the retention window (default_purge=True; held by --uploads-retention-days
```

```
121: uploads_retention_days: float = 0.0,
```

```
126: * ``uploads_retention_days`` guards the upload landing zone (``location=="uploads"``)
```

```
139: window = uploads_retention_days if e.location == "uploads" else older_than_days
```

```
406: "--uploads-retention-days",
```

```
444: uploads_retention_days=args.uploads_retention_days,
```

```
456: "uploads_retention_days": args.uploads_retention_days,
```

```
477: f"{args.uploads_retention_days:g}d) ? run `python -m tools.cleanup_caches`
```

```
to review, ``--apply`` to delete."
```



```

485:      f" uploads ({uploads_dir}): purge source videos > {args.uploads_retention_days:g}d
old"
486:      + ("\" if args.uploads_retention_days > 0 else " [OFF]")

```

15. user (2026-07-02T03:18:23.363Z)

```

335:  # contain upload_dir or /api/process will reject the uploaded paths.
336:  upload_dir: str = "output/uploads"
deploy/cloudrun:
Dockerfile
README.md
cloudbuild.yaml

deploy/digitalocean:
README.md
bootstrap.sh
gpu_setup.sh
mount_scratch.sh

```

16. user (2026-07-02T03:18:35.973Z)

(Bash completed with no output)

17. user (2026-07-02T03:18:37.601Z)

```

DEPLOY_ALPHA.md
GPU_VM_SERVING.md
cloudflared.config.example.yml
droplet
stage_from_bucket.sh

```

18. user (2026-07-02T03:18:44.878Z)

Deploy KhelSutra Studio for alpha ? a shareable page, no SSH

```

> **Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22 ?
`deploy/droplet/README.md`; only a rotated Gemini key + a browser OTP login remain to exercise
the upload?reel CUJ) . **Topology:** split ? SPA on Cloudflare Pages + engine on an `api.`
cloudflared tunnel, Cloudflare Access over both (the engine's `HOSTING_PLAN.md` $2 alpha plan).
. **Cost:** ~$0/mo standing (Pages + Tunnel + Access are free ?50 users) + Gemini per-run
(~$0.05/match).

```

This turns the local-only flow (run engine + SPA on your box, drive it yourself) into a **page** you can share with alpha testers ? they open `app.khelsutra.guru`, sign in (email allowlist), point at a video, and get a reel. No SSH, no home IP exposed, no inbound ports.

```

> **Outgrown the droplet?** To move the live engine onto a GCP L4 GPU VM (more CPU/RAM for
ffmpeg
> + optional WASB GPU detection) and cut the tunnel over to it, see
**[`GPU_VM_SERVING.md`](GPU_VM_SERVING.md)**
> ? same tunnel + Access, no DNS change, droplet kept as fallback (~$500?600/mo while the VM
runs).

```

0. The shape

```

...
tester browser
  ? https
  ?
Cloudflare edge (free): DNS . TLS . Access (email-OTP allowlist) in front of BOTH hostnames
??? app.khelsutra.guru ? Cloudflare Pages (the web/ SPA; built with VITE_API_BASE)
??? api.khelsutra.guru ? cloudflared Tunnel (outbound-only) ?? 127.0.0.1:8000 on YOUR box
                        ? FastAPI engine

```

```

optional)                                ? (Gemini segmenter; GPU

output/ on disk                           ? async jobs, SQLite,
` ``

```

```

- **No inbound ports.** `cloudflared` dials out; the engine stays bound to `127.0.0.1:8000`. The
box's home/public IP is never exposed.
- **Auth at the edge.** Cloudflare Access gates both hostnames; the engine *also* verifies the
`Cf-Access` JWT in-app (defence in depth ? `security.edge_auth_enabled`), so a direct hit to the
tunnel can't spoof identity.
- **Cross-origin, on purpose.** `app.` (SPA) and `api.` (engine) are different origins ? the SPA
is built with `VITE_API_BASE=https://api.khelsutra.guru/api` (PR #64) and the engine locks CORS
to the SPA origin via `RALLY_CORS_ALLOW_ORIGINS` (no engine code change ? it's already env-
driven).

> **Alternative (single-origin):** if you'd rather run **one box** that serves the SPA *and* the
API behind one origin (Caddy reverse-proxy, no Pages, no cross-origin), see
`LOCAL_APPLIANCE_ARCHITECTURE.md` D1. This runbook ships the **split** topology you chose
(stable Pages-hosted page + a thin API tunnel).

```

```
---
```

1. Prerequisites

```

- A **box** to run the engine: the CPU droplet `168.144.126.176` (Gemini-only ? no GPU; fine for
alpha) **or** your Windows+WSL GPU box (real WASB). `ffmpeg` + Python 3.8+ installed; the
`badminton-highlight-indexer` engine checked out.
- A **Cloudflare account** with the `khelsutra.guru` zone (you already run the marketing site
there).
- `cloudflared` installed on the box (`https://developers.cloudflare.com/cloudflare-
one/connections/connect-networks/downloads/`).
- A **fresh** `GEMINI_API_KEY` (rotate the chat-shared one ? it uploads downsized chunks to
Google; this is a paid + cloud action).

```

```
---
```

2. Engine (the `api.` side)

2.1 Install (with the auth extra)

```

` `` bash
cd badminton-highlight-indexer
pip install -e .[auth]          # PyJWT[crypto] ? REQUIRED when edge auth is on, else every
authenticated request is rejected 401
` ``

```

2.2 The safe-exposed config

```

Create / edit `config.json` on the box. **Where it must live:** the engine reads
`$RALLY_APP_HOME/config.json` if `RALLY_APP_HOME` is set, else `./config.json` relative to *the
directory you launch the server from* ? **not** necessarily the repo root. If the engine reads a
different (or absent) file it silently falls back to all-defaults, which means **edge auth is
OFF**. To make writer and reader agree by construction, write it with `python -m backend.main
--setup` (it targets the exact path the server reads) and edit *that* file:
` `` jsonc
{
  "security": {
    "edge_auth_enabled": true,                // turn the CF-Access JWT verify ON
    "allowed_video_root": "/srv/khelsutra/incoming", // the path-jail ? REQUIRED when exposed
    "upload_dir": "/srv/khelsutra/incoming/uploads", // MUST sit UNDER allowed_video_root (see
$7)
    "cf_team_domain": "https://<your-team>.cloudflareaccess.com",
    "cf_access_aud": "<the CF Access application AUD>" // filled in $4
  },
}
=====

```

README.md
config.json.example
engine.env.example
khelsutra-engine.service

19. user (2026-07-02T03:18:46.326Z)

```
__init__.py
__pycache__
auth.py
job_worker.py
static
uploads.py
5:under security.upload_dir, then submitted to /api/process by the returned path.
7:aborts partial uploads and the client re-inits. Per-user partitioning of upload_dir
50:     upload_dir = getattr(sec, "upload_dir", None) or "output/uploads"
57:     return upload_dir, max_bytes, part_bytes, exts
71:def upload_init(req: UploadInitRequest):
73:     upload_dir, max_bytes, part_bytes, exts = _upload_cfg()
89:     os.makedirs(upload_dir, exist_ok=True)
93:     require_free_bytes(upload_dir, req.total_size_bytes, stage="upload")
95:     tmp_path = os.path.join(upload_dir, f".partial-{upload_id}")
112:async def upload_part(upload_id: str, index: int, request: Request):
158:def upload_complete(upload_id: str, req: UploadCompleteRequest):
160:     upload_dir, _, _ = _upload_cfg()
170:     final_path = os.path.join(upload_dir, st["filename"])
175:     final_path = os.path.join(upload_dir, f"{stem}-{upload_id[:8]}{ext}")
189:def upload_abort(upload_id: str):
```

20. user (2026-07-02T03:19:00.485Z)

```
====
    ap.add_argument(
        "--uploads-dir",
        default=None,
        help="upload landing zone to sweep (default <output-dir>/uploads; match
security.upload_dir)",
    )
    ap.add_argument(
        "--uploads-retention-days",
        type=float,
        default=7.0,
        help="sweep uploaded source videos older than N days (default 7; 0=off). The T11 privacy
window.",
    )
    ap.add_argument(
        "--summary",
        action="store_true",
        help="print a single-line nudge (only if reclaimable >= --threshold-gb); for hooks",
    )
```

21. user (2026-07-02T03:19:10.797Z)

```
backend/pipeline/detectors/base.py:162:         f"Free space (e.g. python -m
tools.cleanup_caches) and retry."
README.md:171:*     **Manual sweep (dry-run-first):** `tools/cleanup_caches.py` reports and (with
`--apply`) reclaims what failed/old runs left behind, across both the WSL stage dir and
`output/`:
README.md:173:     python -m tools.cleanup_caches                # report what's reclaimable
(safe; deletes nothing)
README.md:174:     python -m tools.cleanup_caches --apply          # delete the regenerable
caches
README.md:175:     python -m tools.cleanup_caches --keep-video GX010125  # protect one video's
cache (id or filename stem)
```

README.md:176: python -m tools.cleanup_caches --older-than 7 # only purge entries older than 7 days

README.md:249:* `indexing.wasb.keep\_frames` / `indexing.tracknet.keep\_frames` ? after a **successful** trajectory-detector run, the decoded frame cache and the staged video copy (the dominant disk consumer ? ~GBs per 4K video) are deleted automatically. Default `false` (= auto-delete); set `true` only for debugging. On **failure** they're always kept so a re-run resumes. `indexing.{wasb,tracknet}.min\_free\_gb` warns before a run if WSL free space is below the threshold (0 = off). See **§ Cache hygiene** below and `python -m tools.cleanup\_caches`.

README.md:443: cleanup_caches.py # Dry-run-first cache GC (WSL ~/clips + output/); see "Cache Hygiene" above

docs/CODE_MAP.md:30:| **Ops / maintenance tool** (not part of the served app ? cache GC, batch admin) | `tools/` | Run as `python -m tools.<name>` (it's an importable package). e.g. `tools/cleanup\_caches.py`. Keep the destructive decision path pure + unit-tested. |

docs/CODE_MAP.md:195:- `@tools/cleanup\_caches.py` ? **top-level** `tools/` ops tool (NOT backend runtime). Dry-run-first cache GC. Pure `::classify`(name,is_dir,location)/`::plan\_deletions` (unit-tested = the destructive decision path) + I/O `::scan\_wsl` (du+find; ? arg-globbing NOT a `for` loop ? the login shell empties loop vars) / `::scan\_windows` / `::\_delete`. CLI: `python -m tools.cleanup\_caches` (report) . `--apply` / `--keep-video <id|stem>` / `--older-than N` / `--include-wasbcache` / `--summary` (hook nudge). Purges `\*\_frames`/staged/proxy/handoff/tmp; keeps CSV/DB/reels. Companion to the runner self-clean (2.2). See `[[cache-hygiene-policy]]` memory.

22. user (2026-07-02T03:19:59.924Z)

Structured output provided successfully